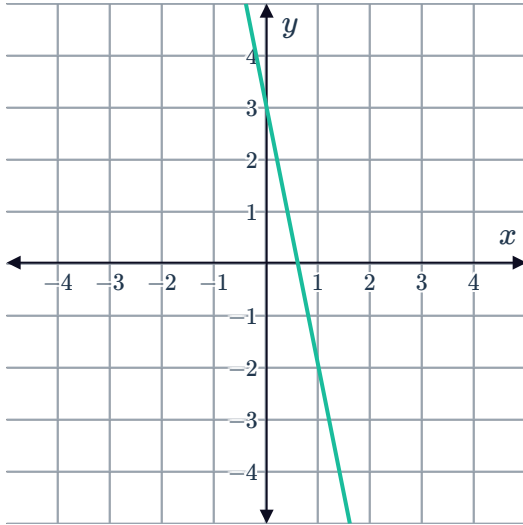


Increasing and decreasing functions

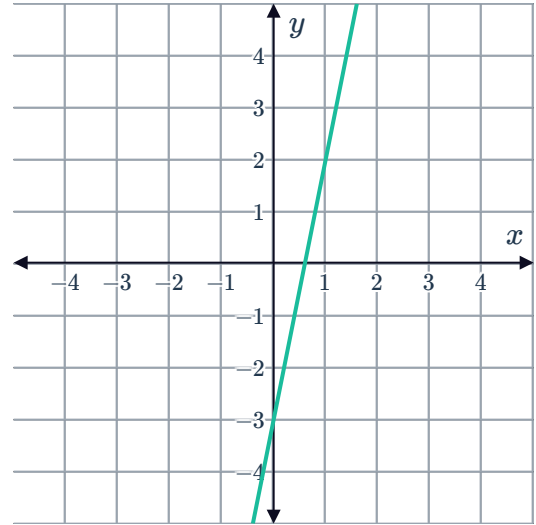


1 State whether each of the following linear functions are increasing or decreasing:

a

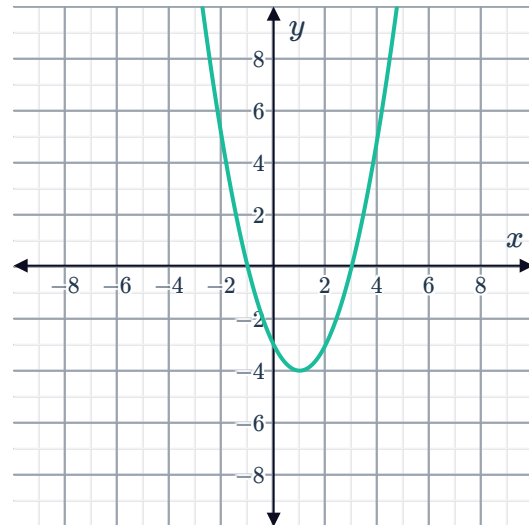


b

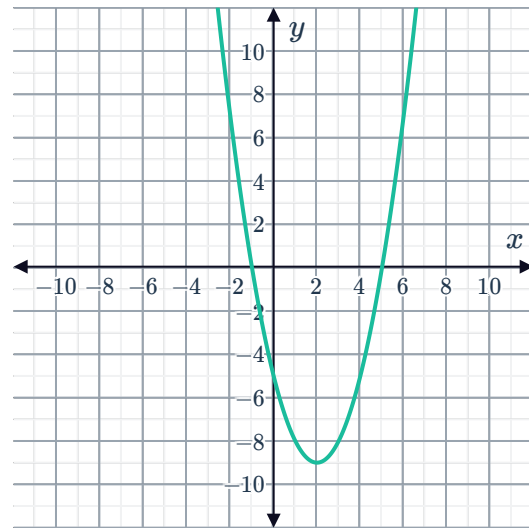


2 Consider the graph of the function $f(x) = (x + 1)(x - 3)$:

- a State the x -value of the point where $f(x)$ changes from decreasing to increasing.
- b State the interval where $f(x)$ is increasing.
- c State the interval where $f(x)$ is decreasing.



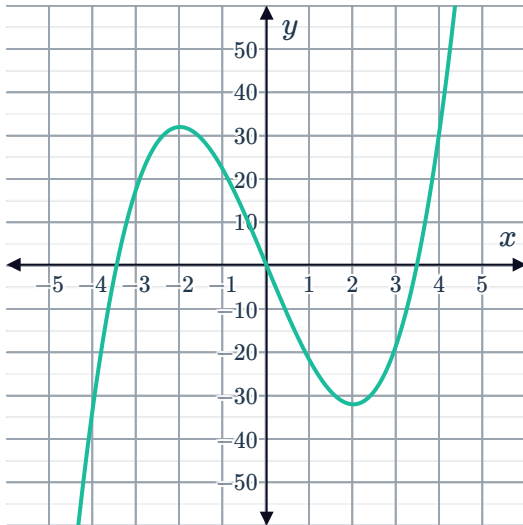
- 3 Consider the given graph. State the interval where $f(x)$ is decreasing.



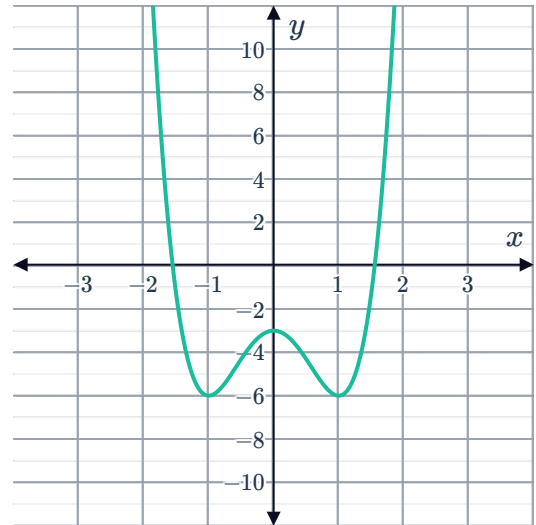
- 4 For each of the following functions:

- i State the interval on which the function is increasing.
- ii State the interval on which the function is decreasing.

a $y = 2x^3 - 24x$



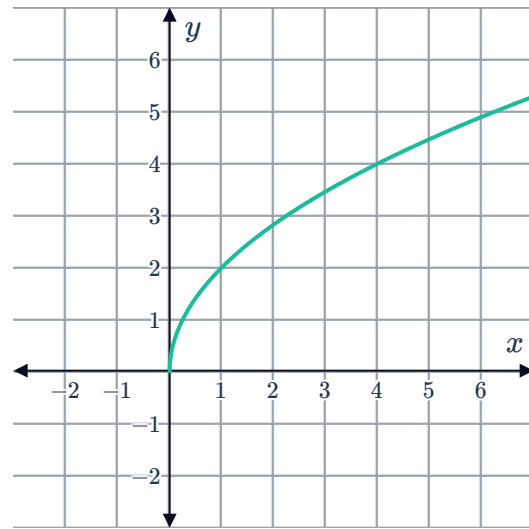
b $y = 3x^4 - 6x^2 - 3$



5 Consider the graph:

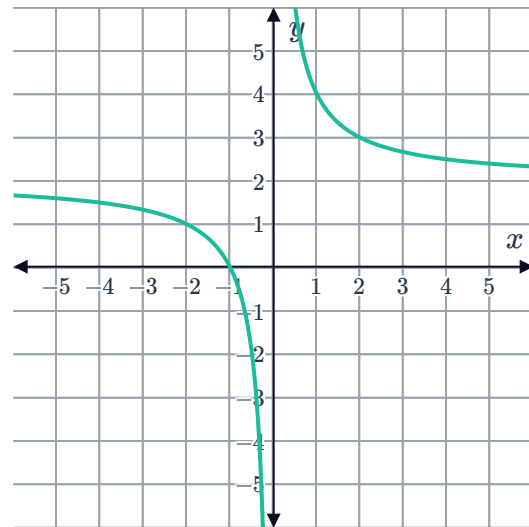


- a State the coordinates of the x -intercept.
- b State the coordinates of the y -intercept.
- c Is the function increasing or decreasing as $x \rightarrow \infty$?



6 Consider the given graph:

- a State the coordinates of the x -intercept.
- b As $x \rightarrow +\infty$, does y increase or decrease?
- c As $x \rightarrow -\infty$, does y increase or decrease?



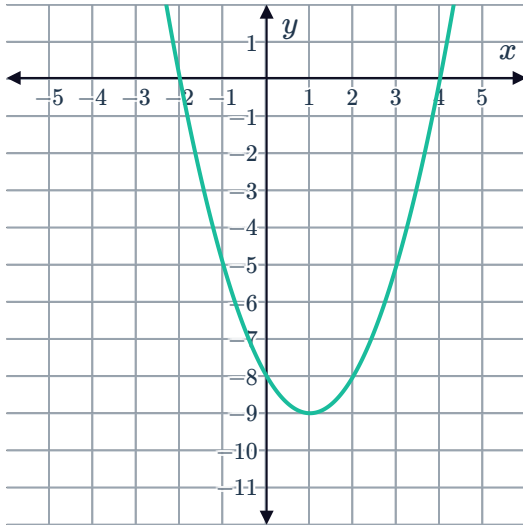


Zeros of functions

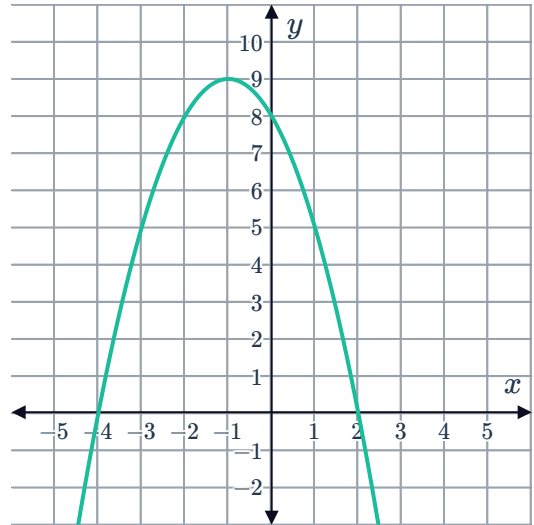
7 For each of the following graphs:

- i Find the coordinates of the x -intercept(s).
- ii Find the zeros of the function.

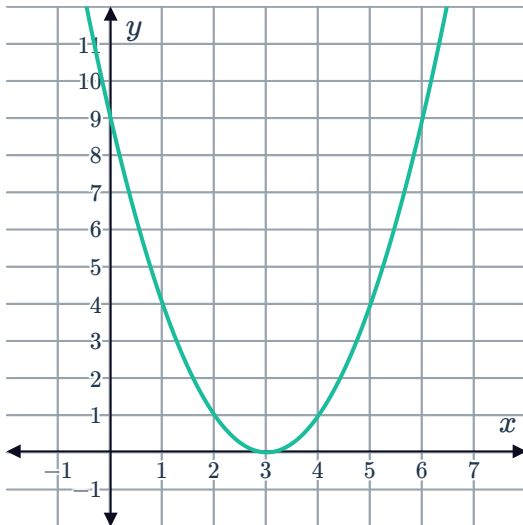
a



b



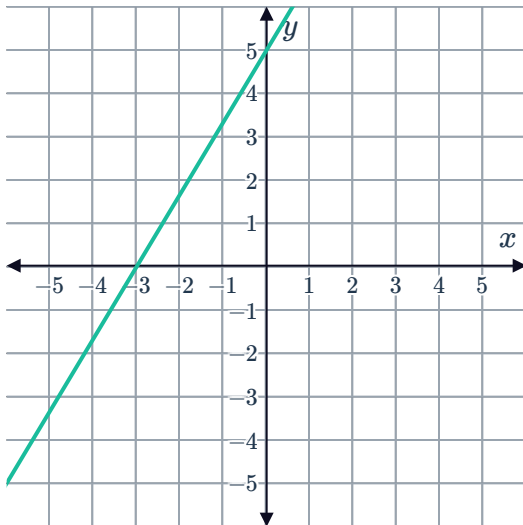
c



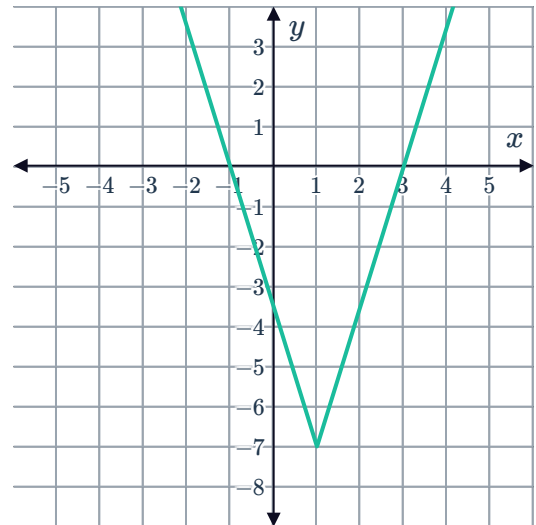
8 Find the zeros of the following functions:



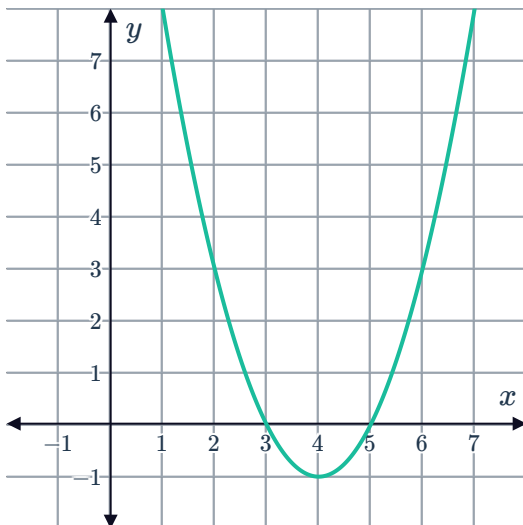
a



b



c

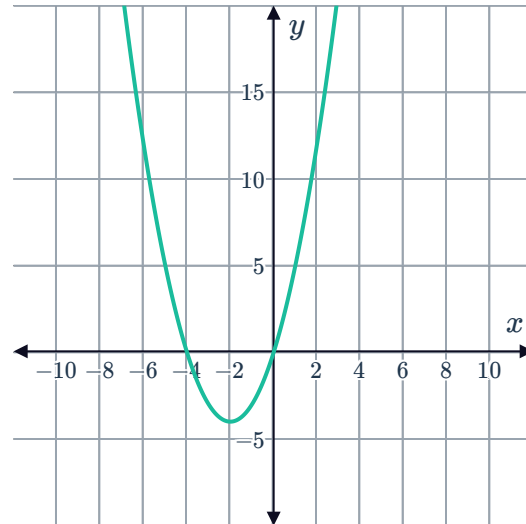




End behavior

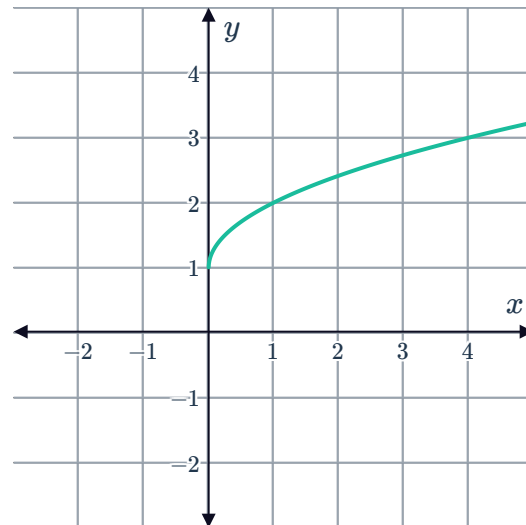
9 Consider the given graph:

- a State the coordinates of the x -intercepts.
- b State the coordinates of the y -intercept.
- c As $x \rightarrow +\infty$, what does y approach?
- d As $x \rightarrow -\infty$, what does y approach?



10 Consider the given graph:

- a State the coordinates of the x -intercepts.
- b State the coordinates of the y -intercept.
- c As $x \rightarrow +\infty$, what does y approach?

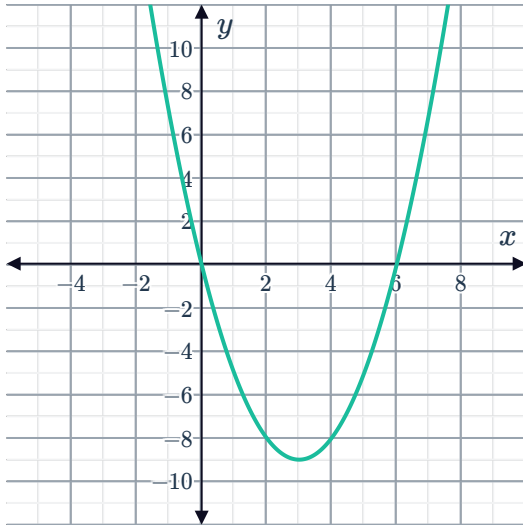


11 For each of the following graphs:

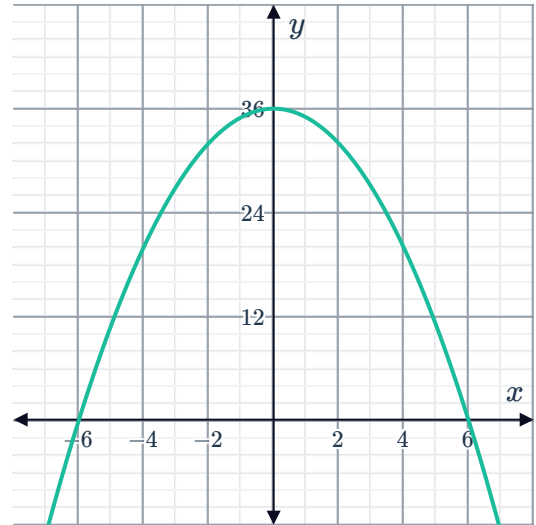


- i State the coordinates of the x -intercepts.
- ii State the coordinates of the y -intercepts.
- iii Describe the behavior of the function as $x \rightarrow \infty$.
- iv Describe the behavior of the function as $x \rightarrow -\infty$.

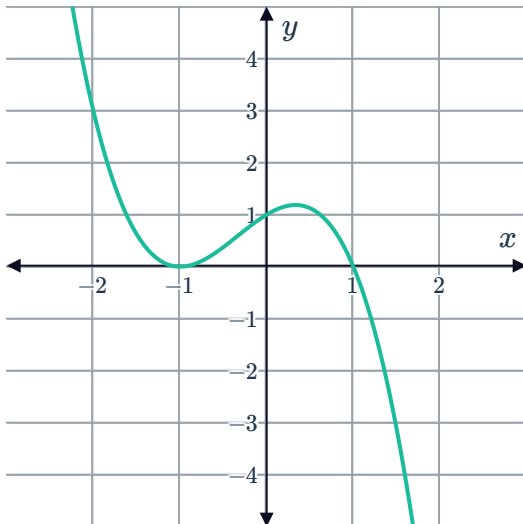
a



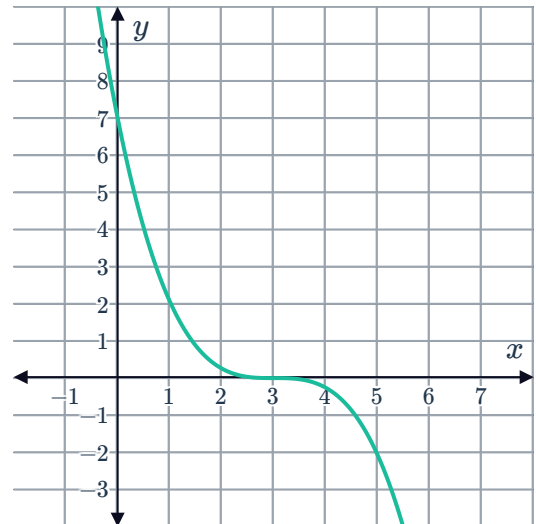
b



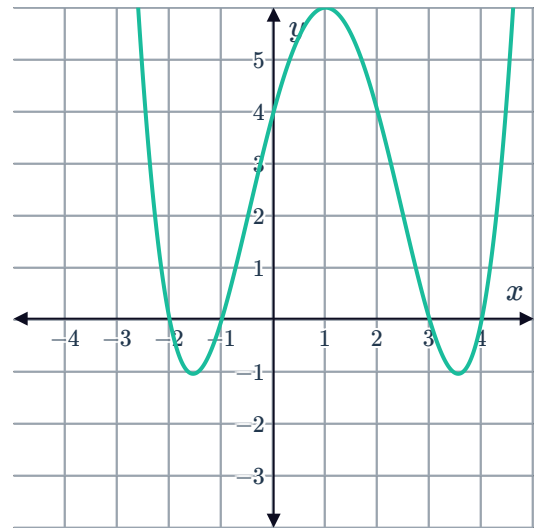
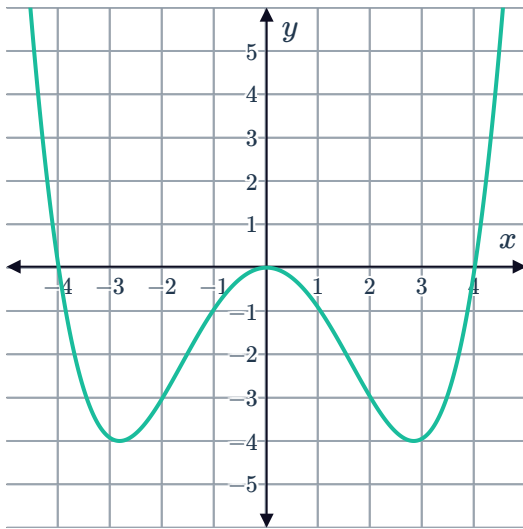
c



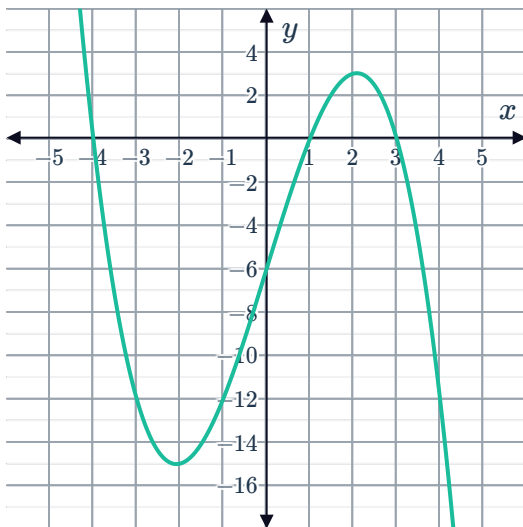
d



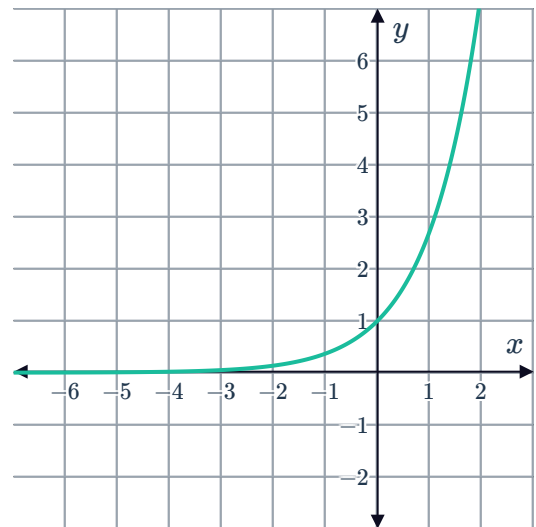
e



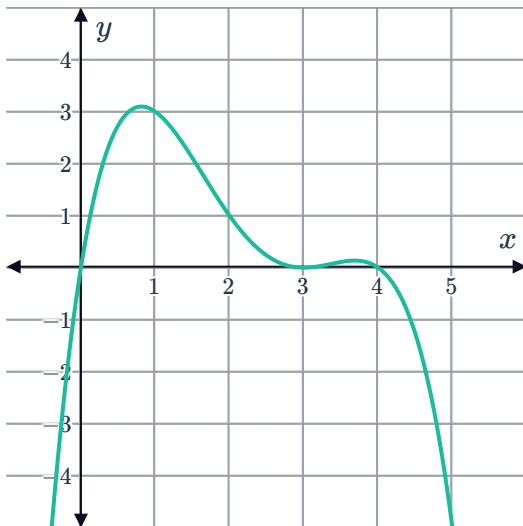
g



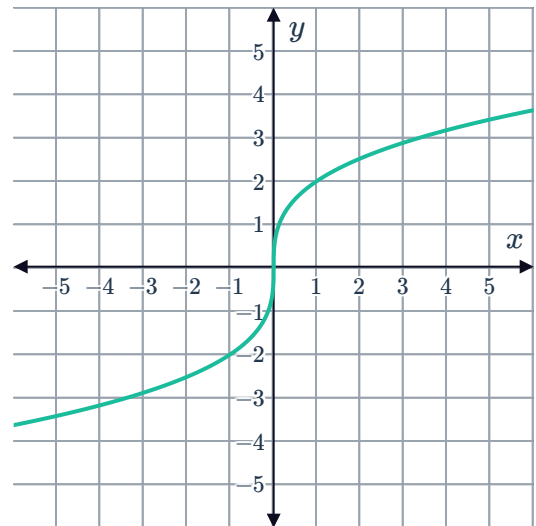
h



i



j

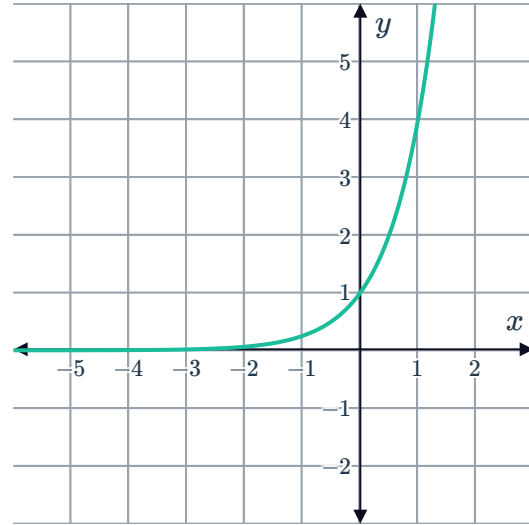




Asymptotes

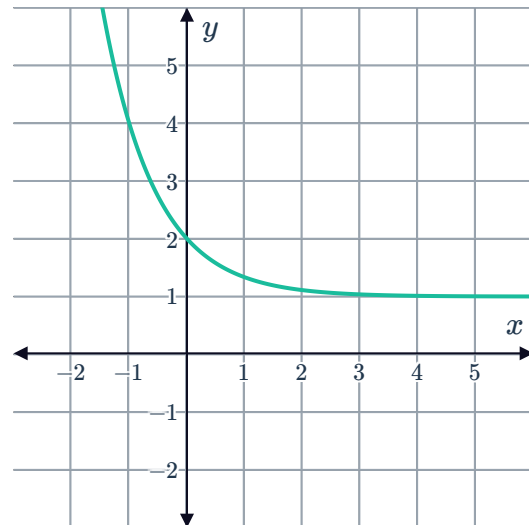
12 A graph of the function $f(x) = 4^x$ is shown below:

- a As $x \rightarrow -\infty$, what does the function approach?
- b State the equation of the horizontal asymptote of $f(x)$
- c True or false: The function is continuous over the interval of all real numbers, $(-\infty, \infty)$.



13 A graph of the function $f(x) = 3^{-x} + 1$ is shown below:

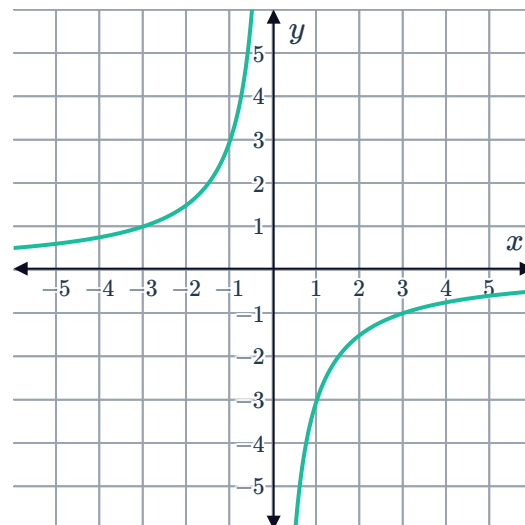
- a As $x \rightarrow \infty$, what does the function approach?
- b State the equation of the horizontal asymptote of $f(x)$
- c True or false: The function is continuous over the interval of all real numbers, $(-\infty, \infty)$.



14 A graph of the function $f(x) = -\frac{3}{x}$ is shown below:



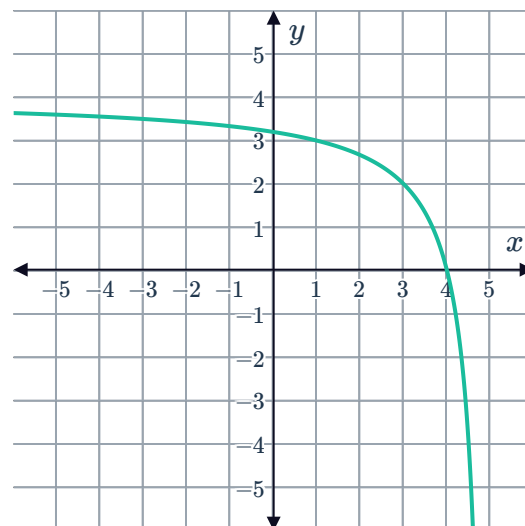
- a Complete the following statements:
- i If x is positive, what value does the function approach as x approaches 0?
 - ii If x is negative, what value does the function approach as x approaches 0?
 - iii As $x \rightarrow \infty$, what does the function approach?
 - iv As $x \rightarrow -\infty$, what does the function approach?



- b State the equation of the vertical asymptote of $f(x)$.
- c State the equation of the horizontal asymptote of $f(x)$.

15 A graph of the function $f(x) = 4 + \frac{4}{x-5}$ is shown below:

- a Complete the following statements:
- i If $x > 5$, what value does the function approach as x approaches 5?
 - ii If $x < 5$, what value does the function approach as x approaches 5?
 - iii As $x \rightarrow \infty$, what does the function approach?
 - iv As $x \rightarrow -\infty$, what does the function approach?



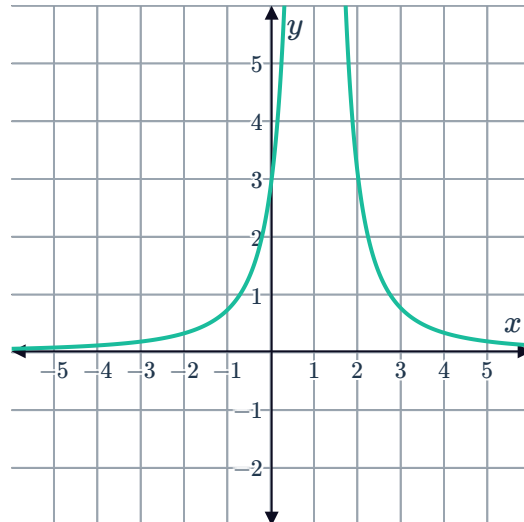
- b State the equation of the vertical asymptote of $f(x)$.
- c State the equation of the horizontal asymptote of $f(x)$.

16 A graph of the function $f(x) = \frac{3}{(x-1)^2}$ is shown below:



a Complete the following statements:

- i If $x > 1$, what value does the function approach as x approaches 1?
- ii If $x < 1$, what value does the function approach as x approaches 1?
- iii As $x \rightarrow \infty$, what does the function approach?
- iv As $x \rightarrow -\infty$, what does the function approach?



- b State the equation of the vertical asymptote of $f(x)$.
- c State the equation of the horizontal asymptote of $f(x)$.

17 The graph of a function $f(x)$ is shown below:

a Complete the following statements:

- i As $x \rightarrow \infty$, what does the function approach?
- ii As $x \rightarrow -\infty$, what does the function approach?

- b State the equation of the horizontal asymptote.
- c State the largest interval of x -values that includes $x = 0$ over which the function is continuous.

